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**Tachizaki**

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(54) **CONNECTING DEVICE**

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**A61M 39/26** (2006.01)

(52) **U.S. Cl.**

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(Continued)

(58) **Field of Classification Search**

CPC ..... **A61M 39/1011**; **A61M 39/26**; **A61M 2039/1083**; **A61M 2039/1088**;

(Continued)

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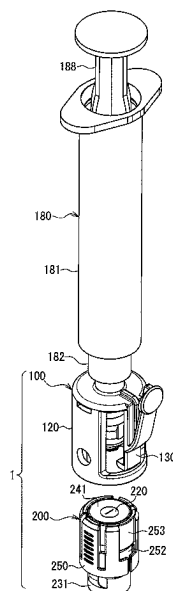
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(57) **ABSTRACT**

A male connector (100) includes a first fitting structure (127) on an inner circumferential surface of a hood (120) surrounding a male member (110). A female connector (200) includes a rotating cylinder (250) rotatably provided on a female connector main body (210) that includes a partition member (220) made of an elastic material. A second fitting structure (257) is provided on an outer circumferential surface of the rotating cylinder. When the male connector is connected to the female connector, the male member passes through a slit (223) of the partition member, the female connector is inserted into the hood, and the first fitting structure fits to the second fitting structure.

**11 Claims, 12 Drawing Sheets**



(52) **U.S. Cl.**

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(58) **Field of Classification Search**

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See application file for complete search history.

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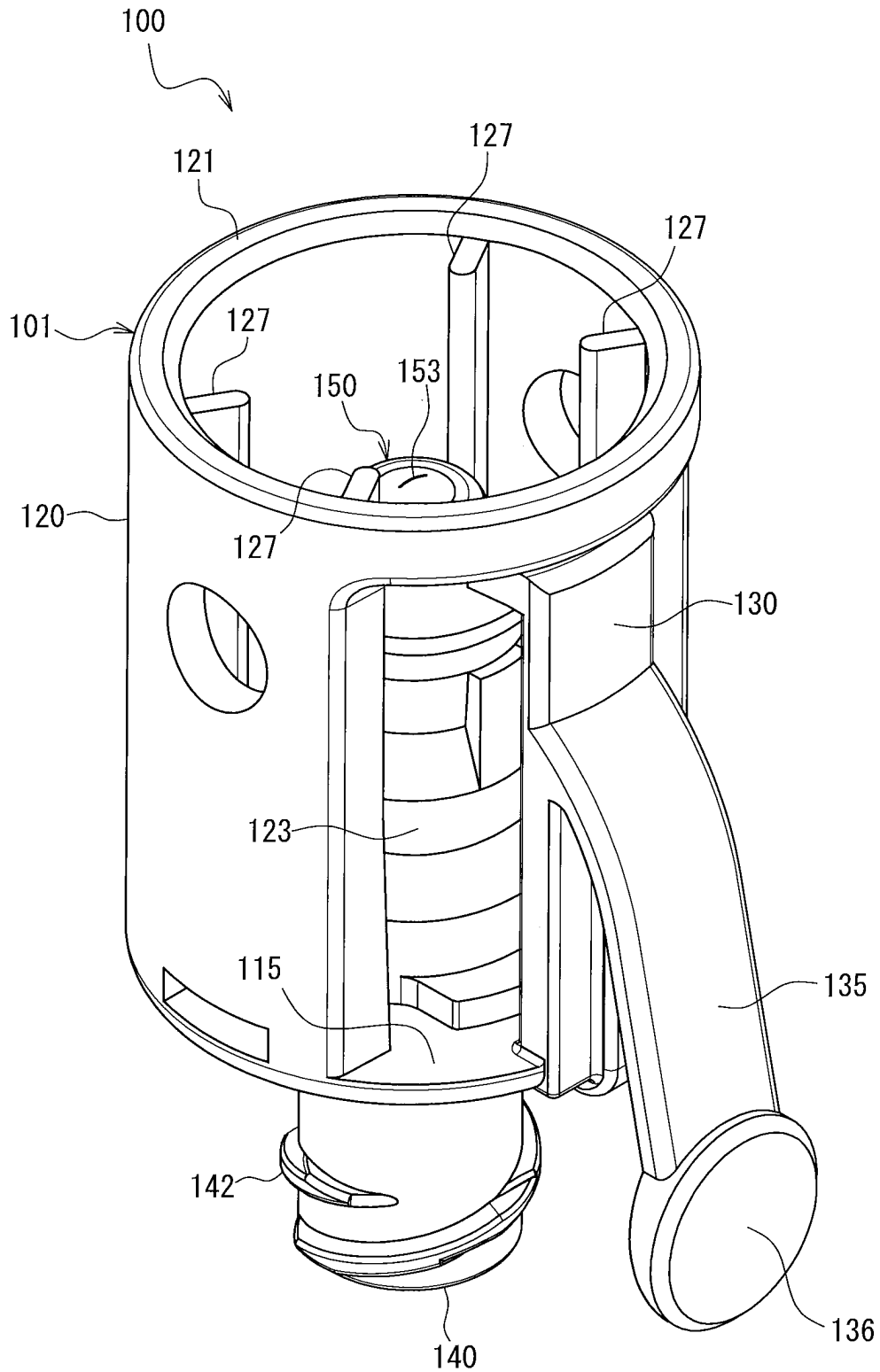


FIG. 1A

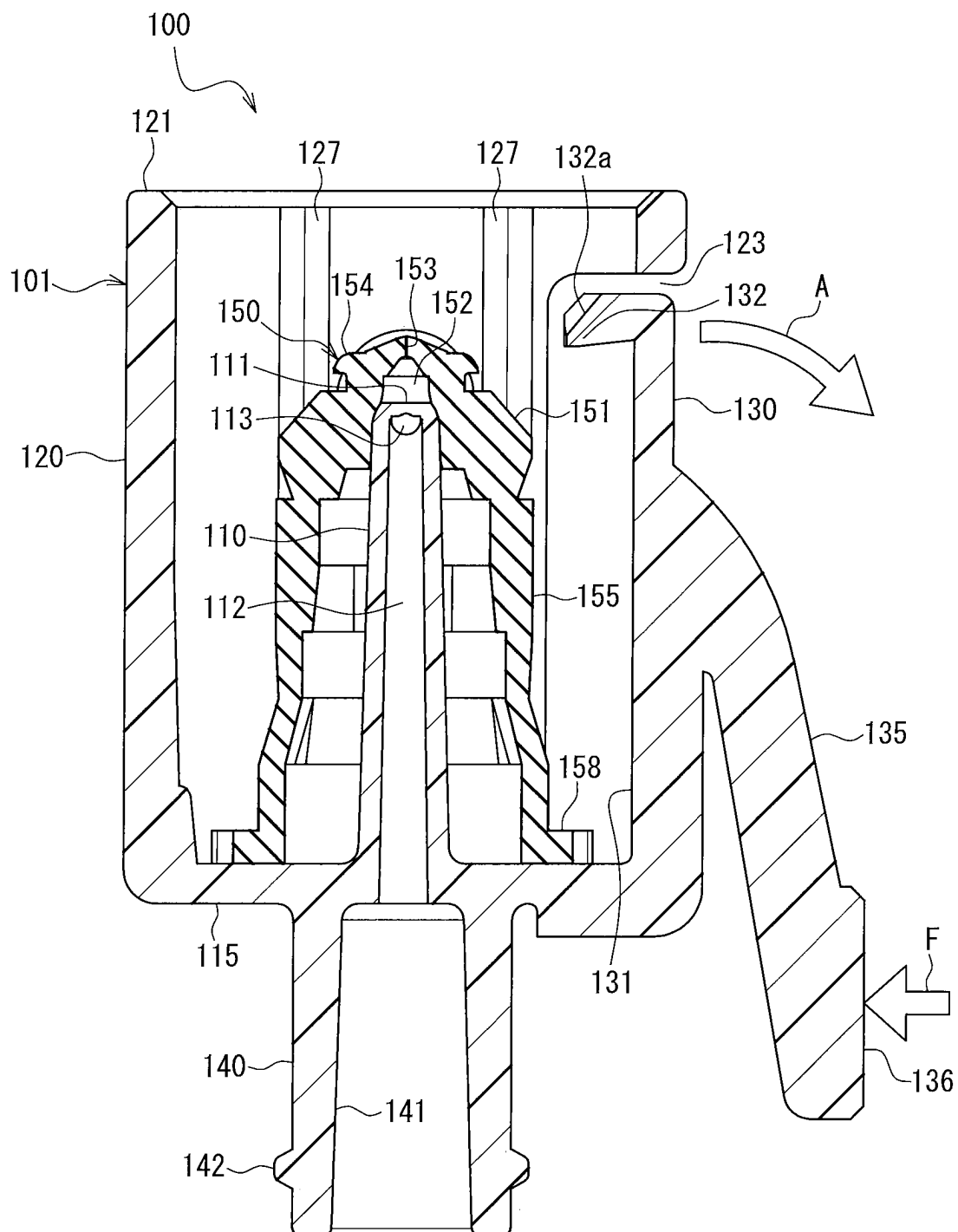


FIG. 1B

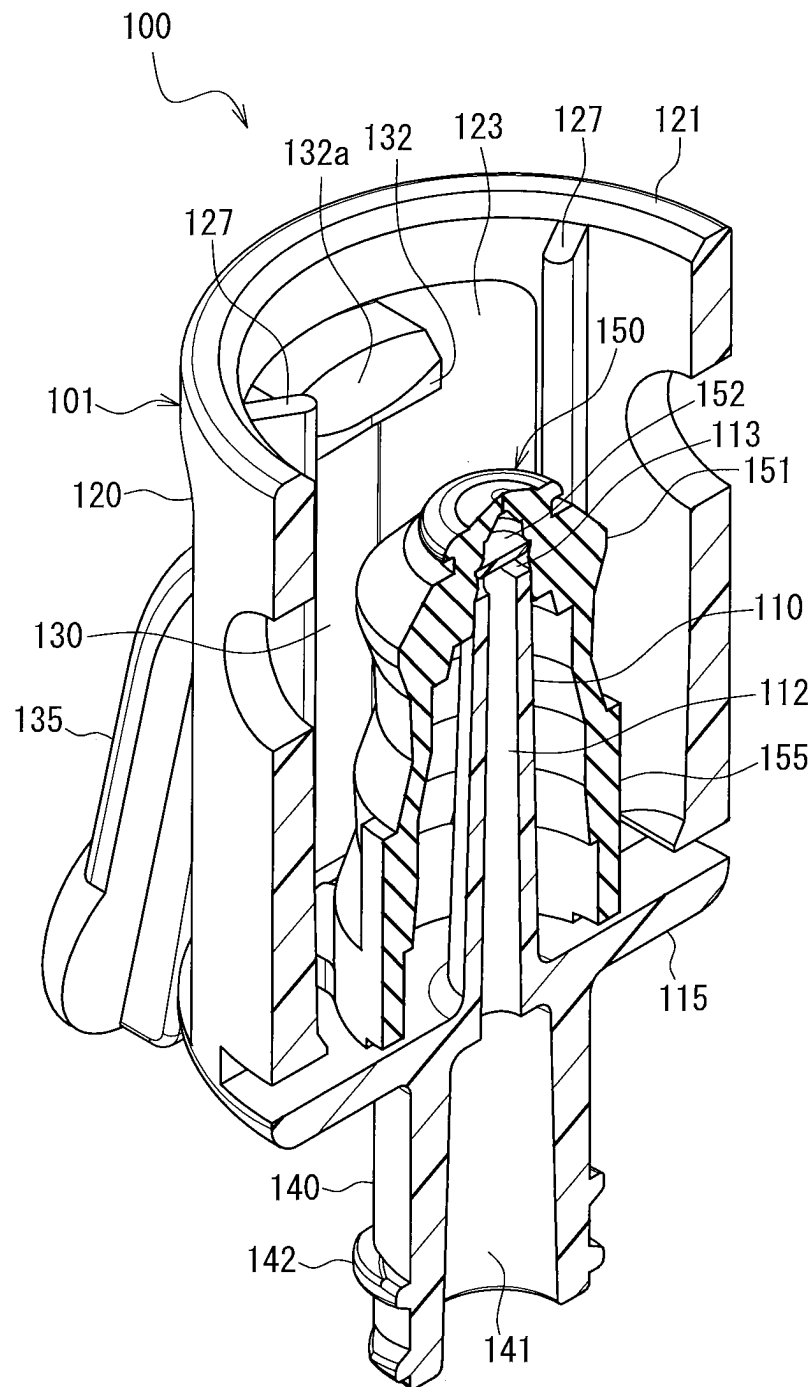


FIG. 1C

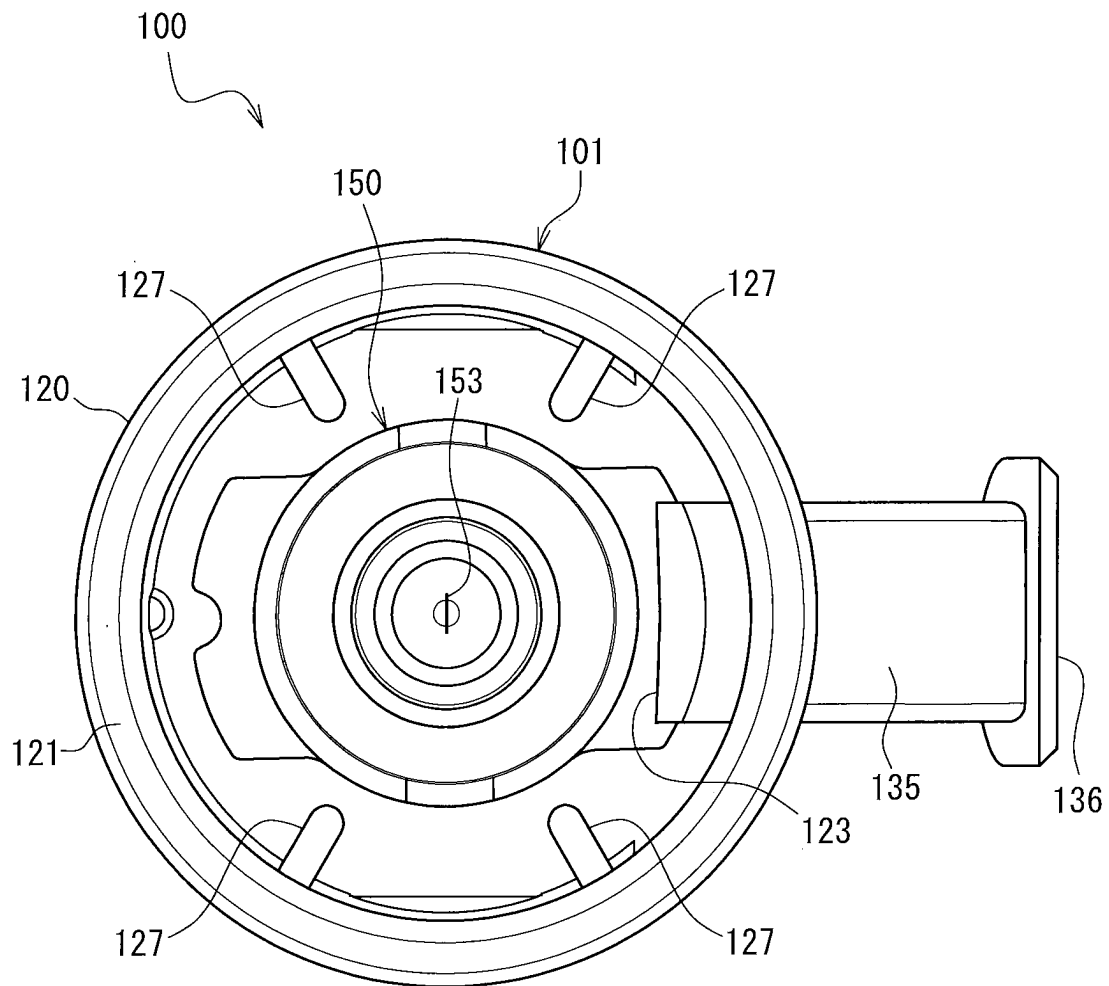


FIG. 1D

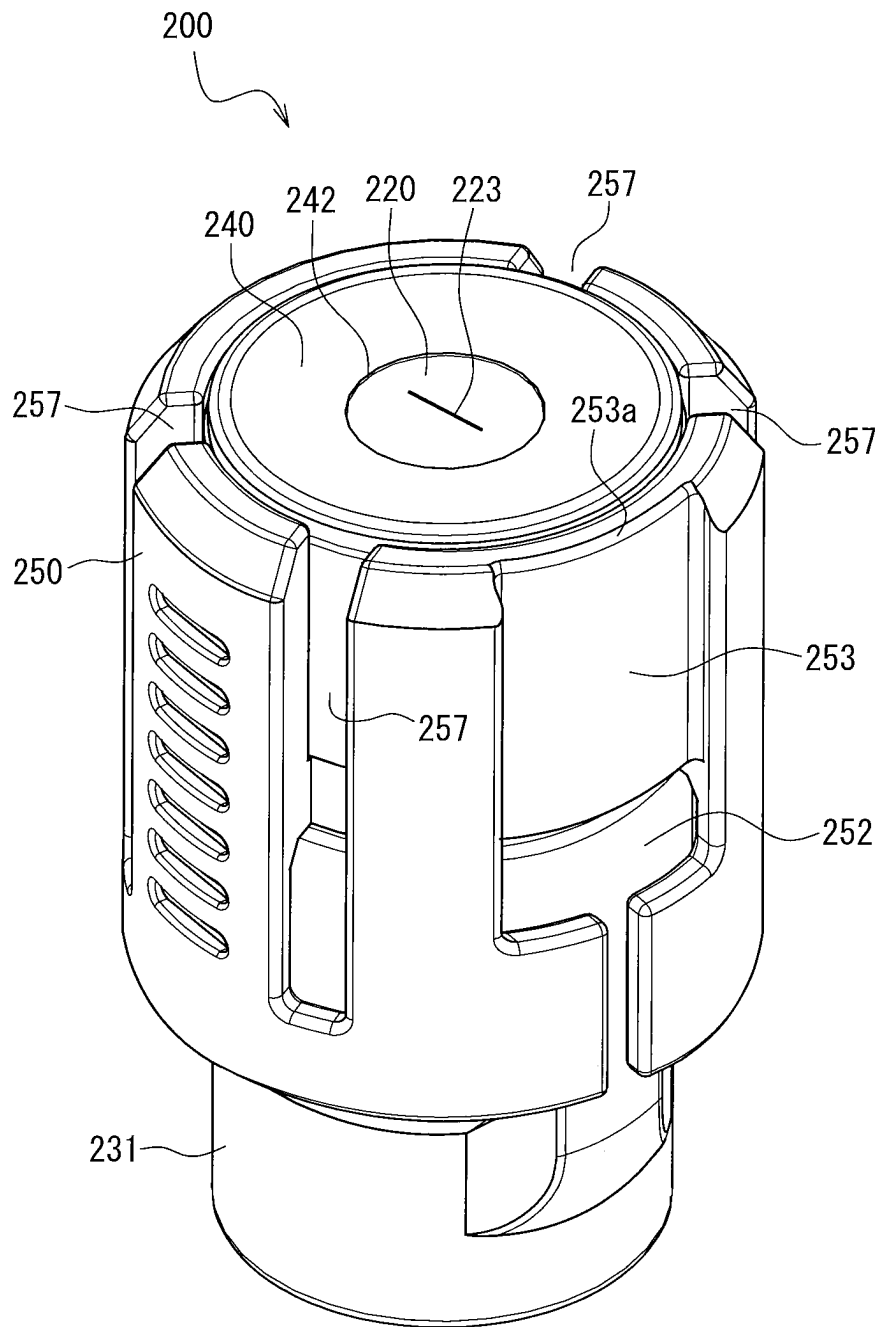


FIG. 2A

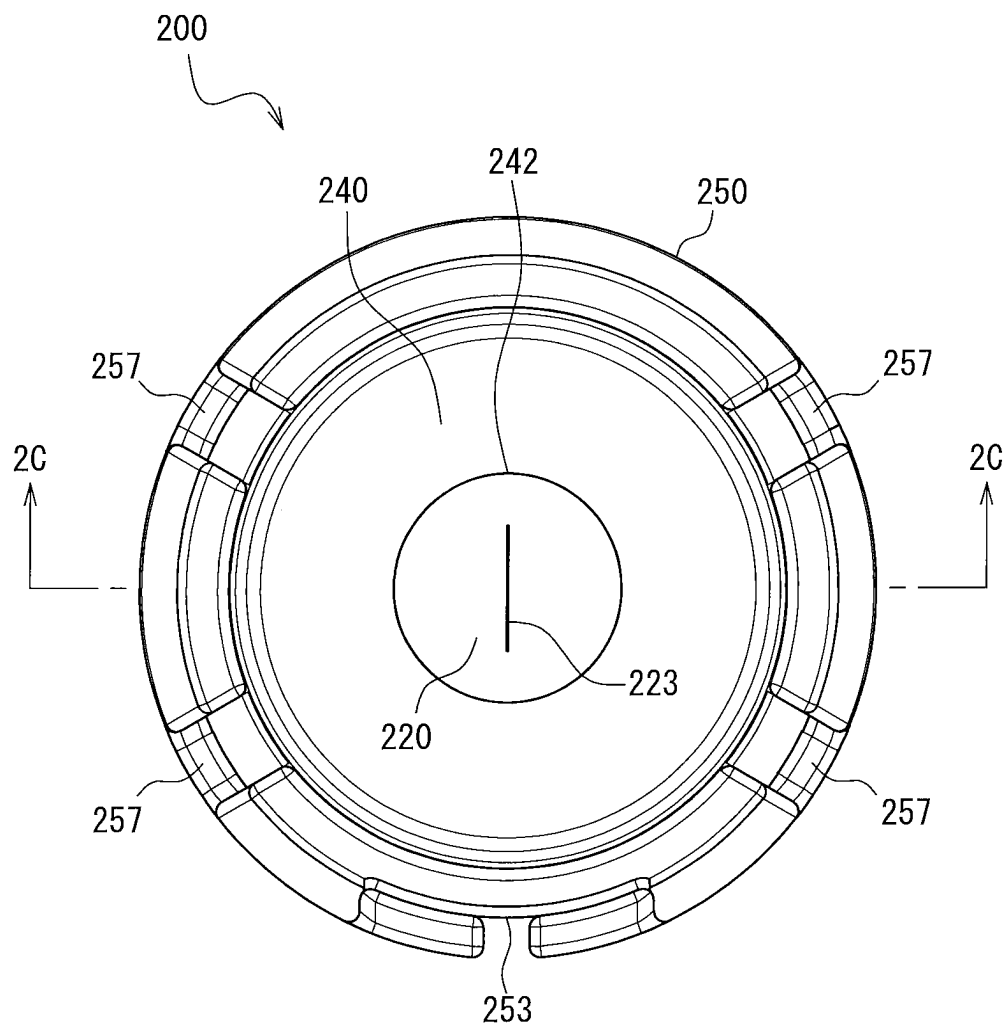


FIG. 2B



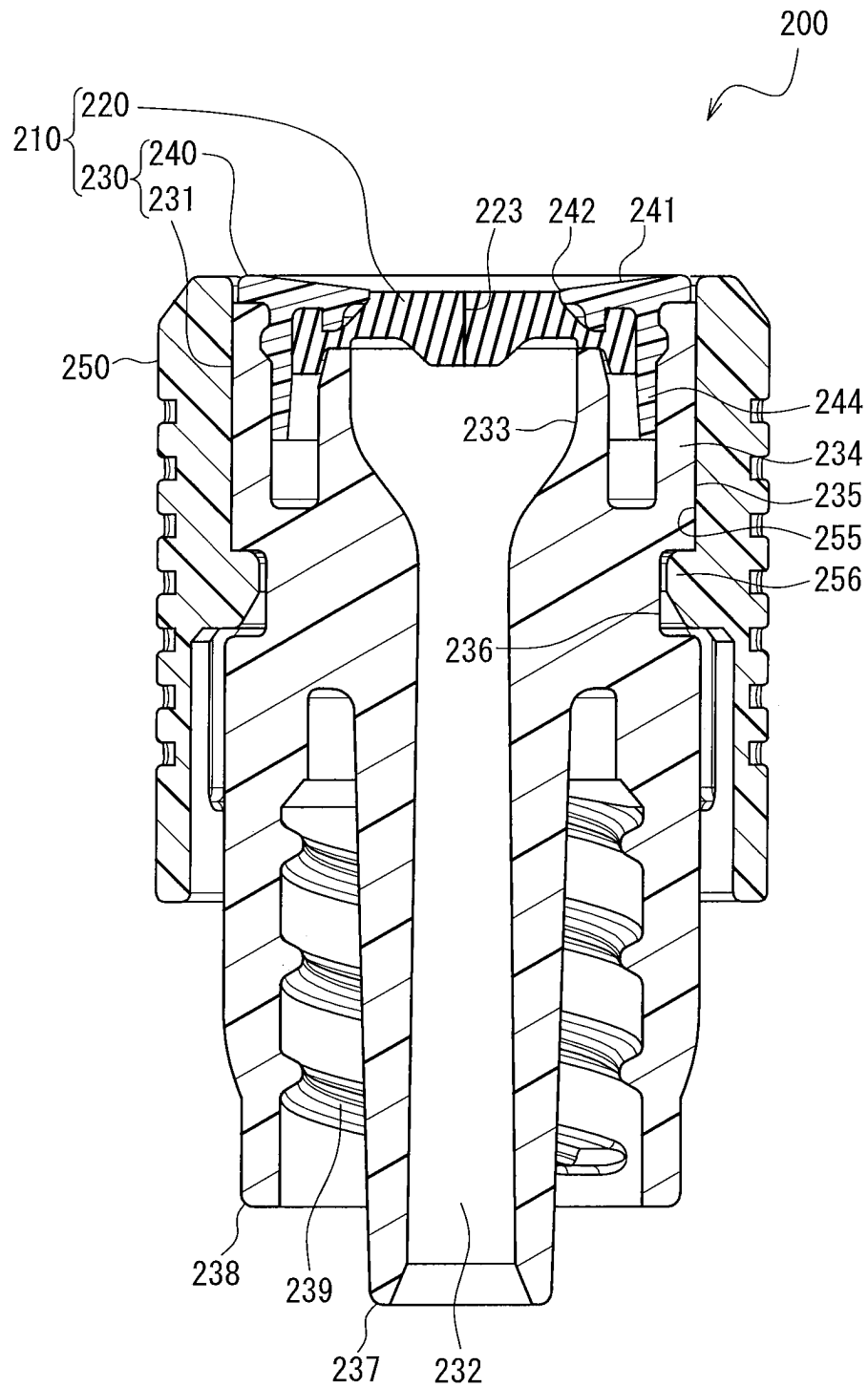


FIG. 2C

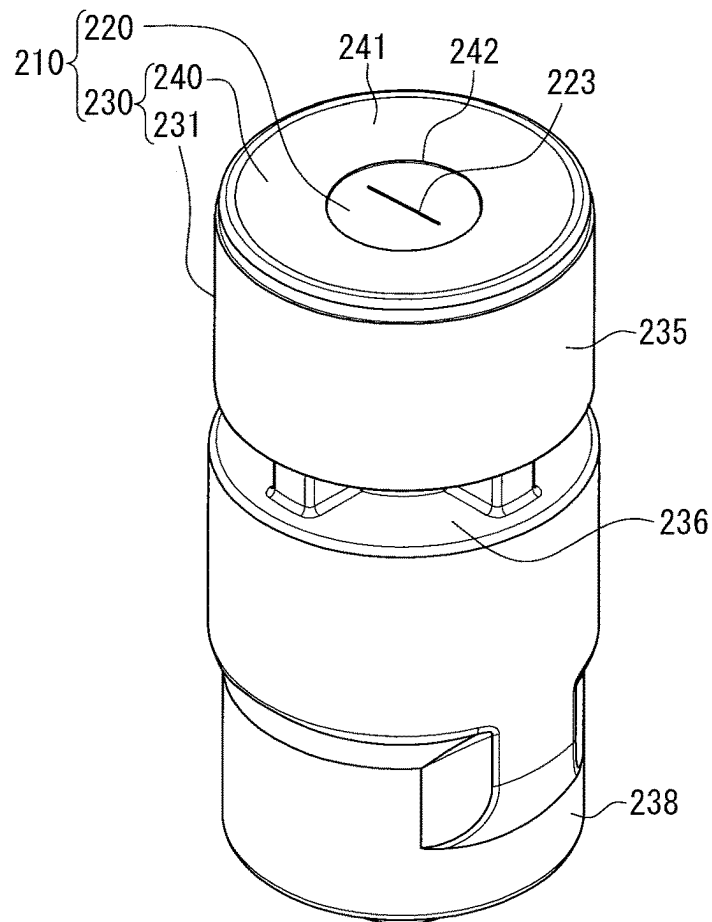
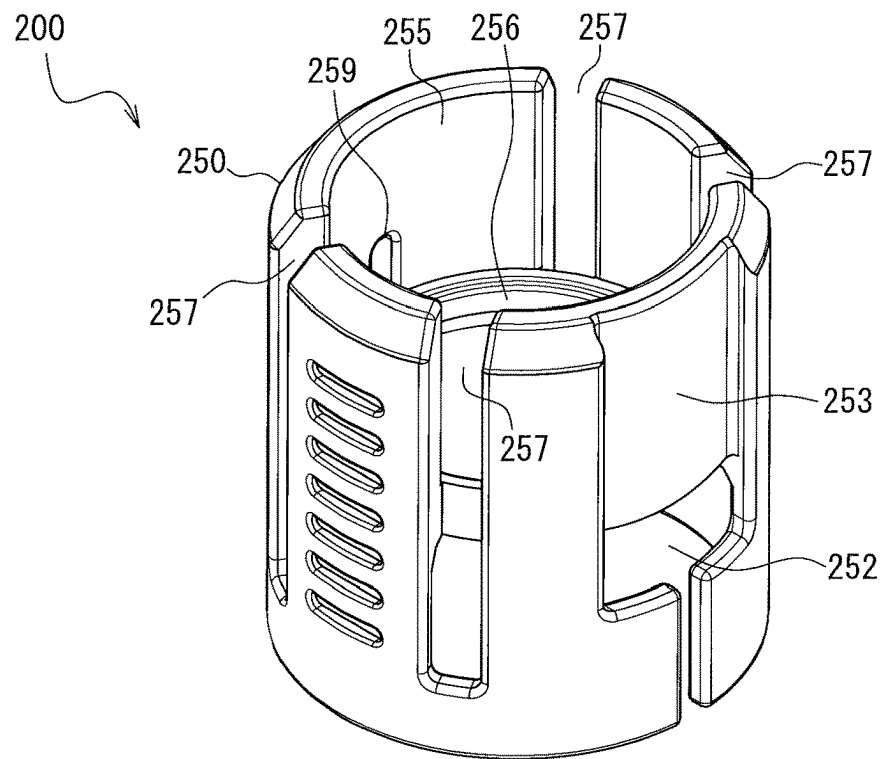


FIG. 2D

FIG. 3A

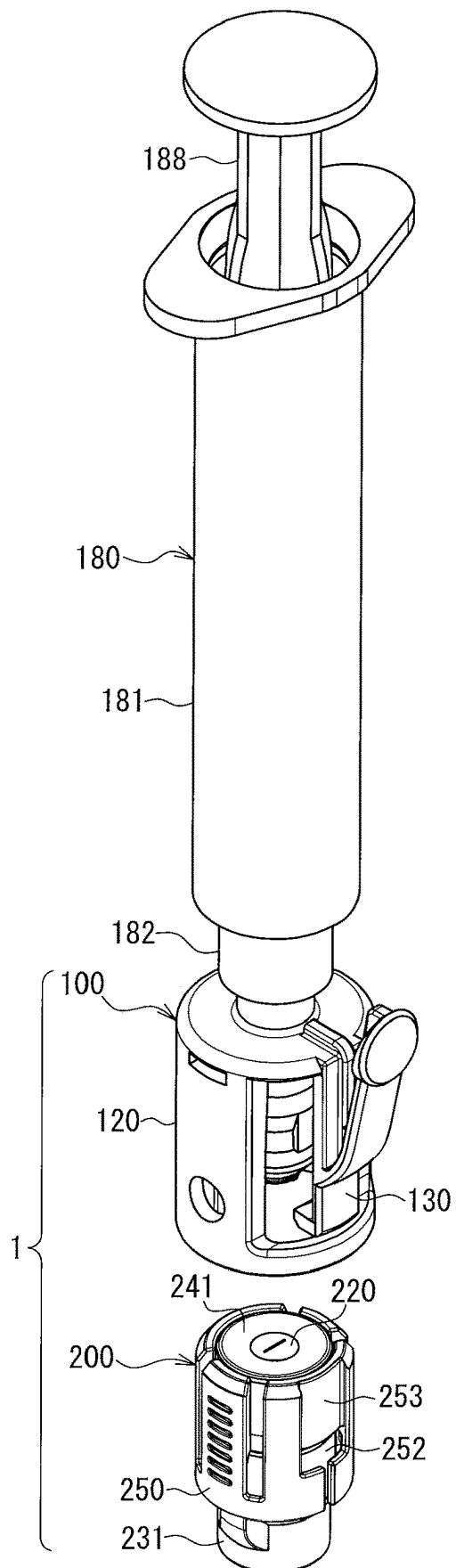


FIG. 3B

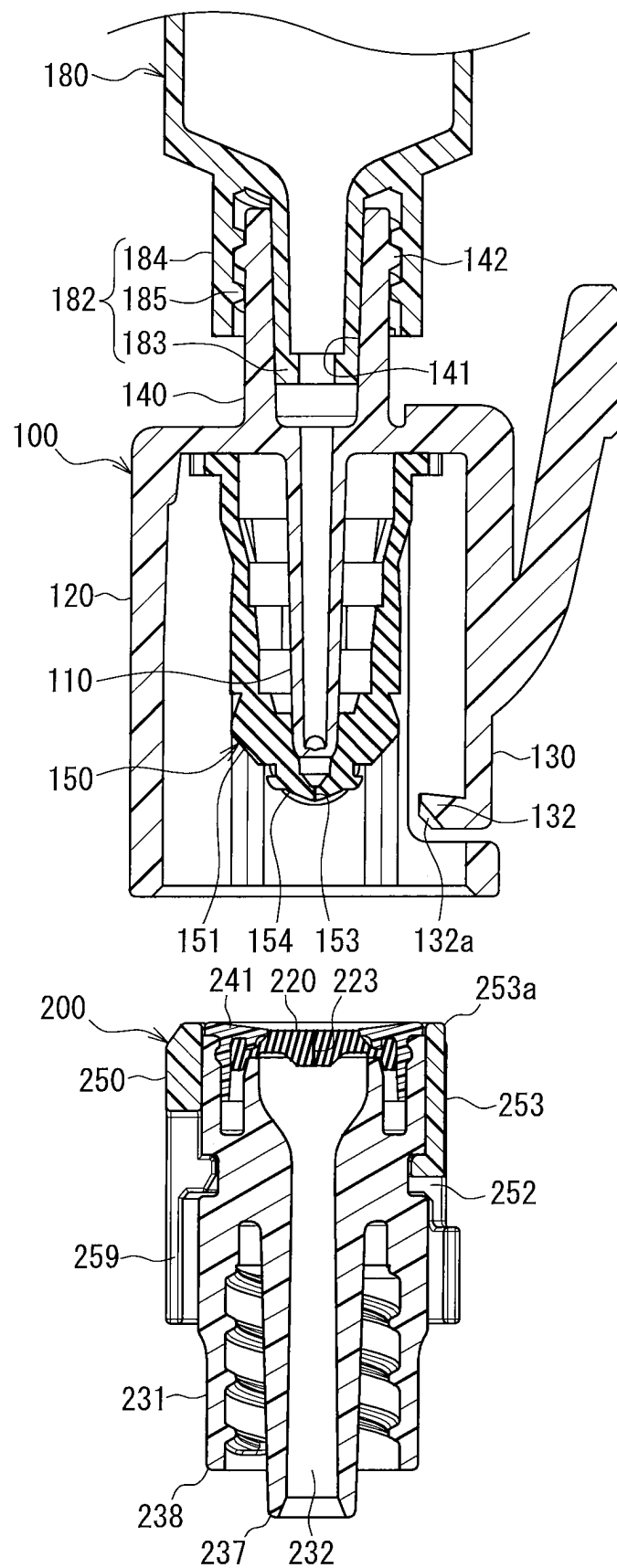
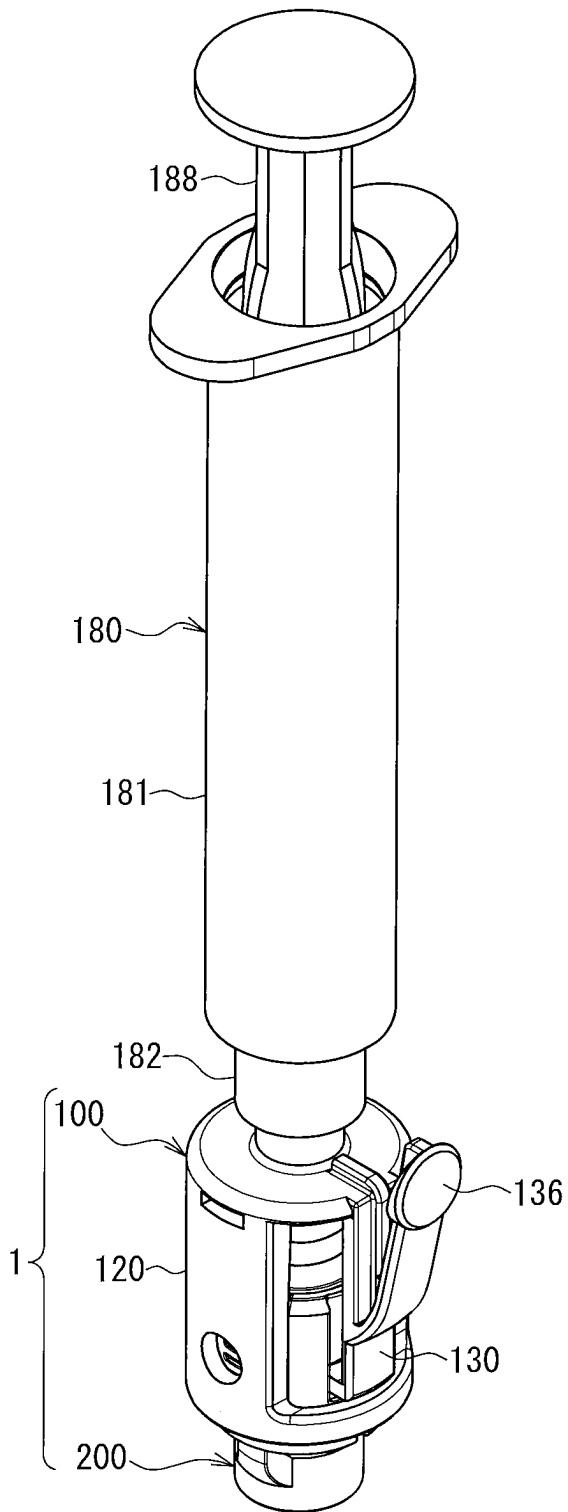
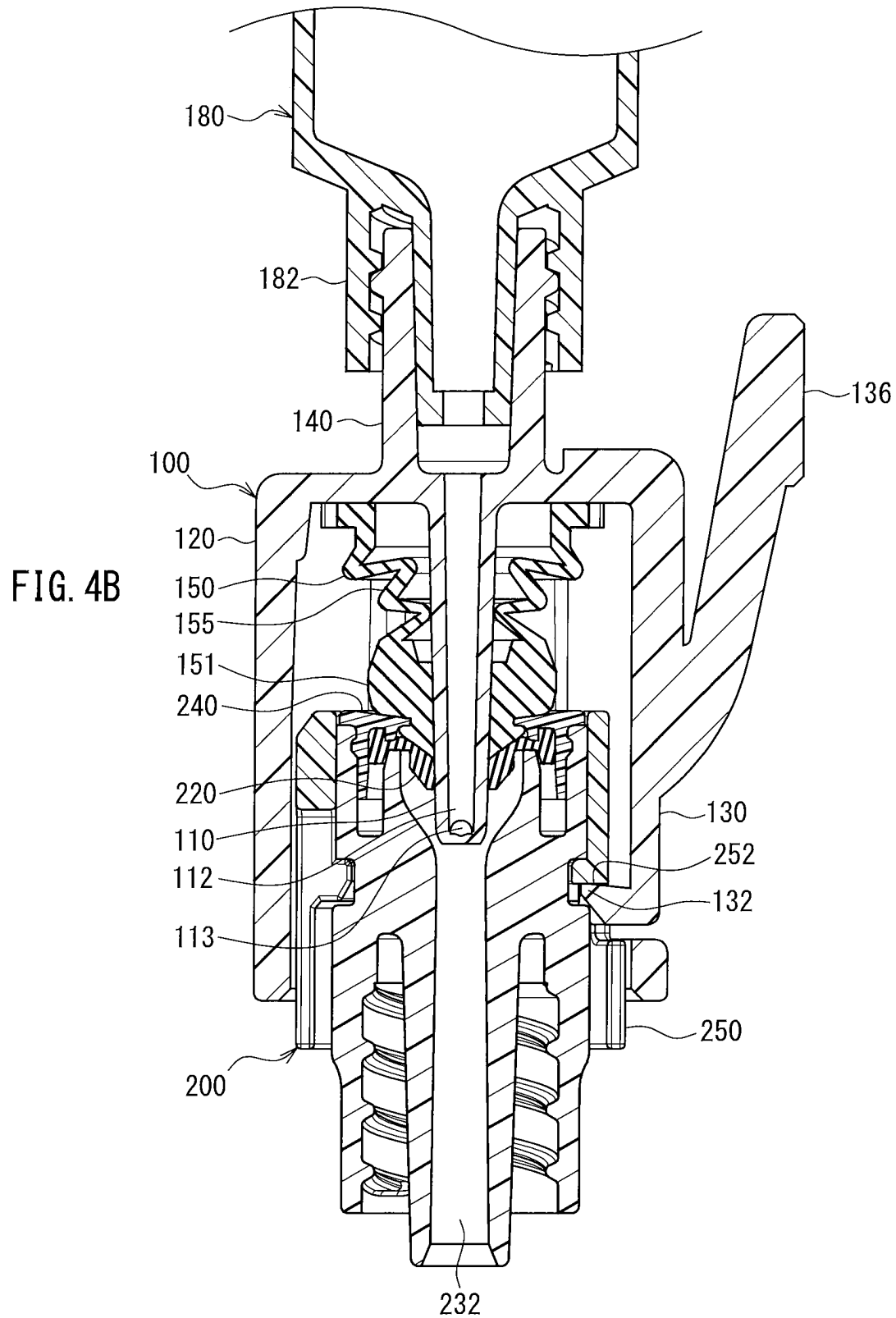


FIG. 4A





# 1

## CONNECTING DEVICE

### TECHNICAL FIELD

The present invention relates to a connecting device 5 constituted by a male connector and a female connector.

### BACKGROUND ART

To administer a drug solution to a patient, first, the prepared drug solution is put into a syringe by suction, and then, the drug solution is administered to the patient through a catheter (or a tube) placed (or inserted) in the patient. The drug solution may contain a hazardous drug such as an anticancer drug. If the hazardous drug leaks to the outside, a worker will be exposed to the drug. In order to prevent drug exposure during drug transfer, it is common to transfer a drug solution using a closed-system connecting device.

Patent Document 1 discloses an example of a closed-system connecting device constituted by a male connector and a female connector.

The male connector of Patent Document 1 includes a rod-shaped male member and a shield that covers the male member. A flow channel through which a drug solution flows is formed within the male member. The flow channel is in communication with the outside via a lateral hole, the lateral hole being formed near a distal end of the male member and extending in the radial direction. The shield is made of an elastic material such as rubber, and a hole through which the male member can pass is formed at a distal end of the shield. In an initial state, an inner circumferential surface of the hole closes the lateral hole of the male member in a liquid-tight manner. When the male connector is connected to the female connector, the shield is compressively deformed in a longitudinal direction of the male connector, and the male member passes through the hole of the shield and protrudes from the shield. When the male connector is disconnected from the female connector, the shield immediately expands to the initial state. The male member is accommodated in the shield, and the lateral hole is closed by the shield.

The female connector of Patent Document 1 includes a self-closing partition member (septum) that is made of an elastic material such as rubber and in which a slit is formed. When the male connector is connected to the female connector, the male member of the male connector is inserted into the slit of the partition member. When the male connector is disconnected from the female connector, the slit immediately returns to its initial state, and the slit is closed in a liquid-tight manner.

The connecting device of Patent Document 1 can be used to transfer a drug solution. For example, it is possible to provide the male connector at a barrel end of a syringe and the female connector at an upstream end of a catheter placed in a patient. It is possible to connect the male connector to the female connector and administer a drug solution stored in the syringe to the patient through the catheter. When the male connector is disconnected from the female connector, the male member is accommodated in the shield, and the slit of the female connector is closed. Therefore, the drug solution is prevented from leaking to the outside from each of the male connector and the female connector.

### CITATION LIST

#### Patent Documents

Patent Document 1: WO 2016/133139A1  
Patent Document 2: WO 2016/152801A1

# 2

## DISCLOSURE OF INVENTION

### Problem to be Solved by the Invention

For example, an anticancer drug may be administered to a vein, bladder, medullary cavity, gastrointestinal tract, or the like, depending on the type and location of the cancer. A catheter is placed in a patient such that its distal end reaches the administration site of the anticancer drug. From the viewpoint of preventing exposure of a worker to the drug, it is desirable to connect a syringe to the catheter via a closed-system connecting device, regardless of where the indwelling catheter is placed.

However, if male and female connectors constituting the connecting device are the same for every site in which a catheter is placed, a misconnection may occur, which is an accident in which, for example, a syringe storing a drug solution prepared for intravenous administration is mistakenly connected to a catheter placed in the bladder.

An object of the present invention is to prevent a misconnection between a male connector and a female connector of a connecting device constituted by the male connector and the female connector, using a simple method.

### Means for Solving Problem

A connecting device of the present invention includes a male connector and a female connector that can be connected to and disconnected from each other. The male connector includes a rod-shaped male member, a hood surrounding the male member, and a first fitting structure provided on an inner circumferential surface of the hood. The female connector includes a female connector main body including a partition member in which a slit is formed, the partition member being made of an elastic material, a rotating cylinder provided on the female connector main body so as to be rotatable around the female connector main body, and a second fitting structure provided on an outer circumferential surface of the rotating cylinder. When the male connector is connected to the female connector, the male member passes through the slit of the partition member, the female connector is inserted into the hood, and the first fitting structure fits to the second fitting structure.

### Effects of the Invention

According to the present invention, when the male connector is connected to the female connector, the first fitting structure fits to the second fitting structure. Therefore, it is possible to prevent a misconnection between the male connector and the female connector using a simple method of employing different configurations for the first fitting structure and the second fitting structure.

The second fitting structure is provided on the rotating cylinder that is rotatable around the female connector main body. Therefore, it is possible to rotate the male connector relative to the female connector main body in a state in which the first fitting structure is fitted to the second fitting structure (i.e., a state in which the male connector is connected to the female connector). Consequently, no rotational force is transmitted between the male connector and the female connector main body.

### BRIEF DESCRIPTION OF DRAWINGS

FIG. 1A is a perspective view of a male connector according to an embodiment of the present invention.

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FIG. 1B is a cross-sectional view of the male connector according to the embodiment of the present invention, taken along a plane containing the central axis of the male connector.

FIG. 1C is a cross-sectional perspective view of the male connector according to the embodiment of the present invention, taken along another plane containing the central axis of the male connector.

FIG. 1D is a plan view of the male connector according to the embodiment of the present invention.

FIG. 2A is a perspective view of a female connector according to the embodiment of the present invention.

FIG. 2B is a plan view of the female connector according to the embodiment of the present invention.

FIG. 2C is a cross-sectional view of the female connector according to the embodiment of the present invention, taken along a plane containing line 2C-2C in FIG. 2B and viewed in the direction of the arrows.

FIG. 2D is an exploded perspective view of the female connector according to the embodiment of the present invention.

FIG. 3A is a perspective view illustrating a method of using a connecting device according to the embodiment of the present invention and shows a state before the male connector is connected to the female connector.

FIG. 3B is a cross-sectional view of main portions in FIG. 3A.

FIG. 4A is a perspective view illustrating the method of using the connecting device according to the embodiment of the present invention and shows a state in which the male connector is connected to the female connector.

FIG. 4B is a cross-sectional view of main portions in FIG. 4A.

### DESCRIPTION OF THE INVENTION

In an aspect of the connecting device of the present invention above, a configuration may be adopted in which the female connector cannot be inserted into the hood when the first fitting structure does not fit to the second fitting structure. This aspect is advantageous in preventing a liquid leakage from the connecting device to the outside when the first fitting structure does not fit to the second fitting structure.

In an aspect of the connecting device of the present invention above, the first fitting structure may be provided nearer to a distal end of the male connector than a distal end of the male member. This aspect is advantageous in preventing a liquid leakage from the connecting device to the outside when the first fitting structure does not fit to the second fitting structure.

In an aspect of the connecting device of the present invention above, the second fitting structure may be provided at or near a distal end of the female connector. This aspect is advantageous in preventing a liquid leakage from the connecting device to the outside when the first fitting structure does not fit to the second fitting structure.

In an aspect of the connecting device of the present invention above, in a process of connecting the male connector to the female connector and in a process of disconnecting the male connector from the female connector, the male connector may be movable in a longitudinal direction of the male member relative to the female connector, with the first fitting structure being fitted to the second fitting structure. This aspect is advantageous in improving the ease of operations for connecting and disconnecting the male connector to and from the female connector.

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In an aspect of the connecting device of the present invention above, one of the first fitting structure and the second fitting structure may be a groove extending in a direction in which the male connector is connected to and disconnected from the female connector. Another one of the first fitting structure and the second fitting structure may be a projection that can fit into the groove. According to this aspect, the first fitting structure and the second fitting structure that can fit to each other when the male connector is connected to the female connector can be realized using a simple configuration.

In an aspect of the connecting device of the present invention above, in a state in which the male connector is connected to the female connector, the male connector may be rotatable relative to the female connector main body. According to this aspect, no rotational force is transmitted between the male connector and the female connector main body.

In an aspect of the connecting device of the present invention above, the male connector may further include a lock lever and a claw provided on the lock lever. The claw may be elastically displaceable in a direction away from the male member. When the male connector is connected to the female connector, the claw may engage with female connector. This aspect is advantageous in preventing unintentional disconnection of the male connector from the female connector.

When the male connector is connected to the female connector, the claw may engage with the rotating cylinder. According to this aspect, in a state in which the male connector is connected to the female connector, no rotational force is transmitted between the male connector and the female connector.

In an aspect of the connecting device of the present invention above, the male connector may further include a shield that is compressively deformable in a longitudinal direction of the male member. In a state in which the male connector is not connected to the female connector, the shield may close an opening of a flow channel provided in the male member. In a state in which the male connector is connected to the female connector, the shield may be compressively deformed in the longitudinal direction of the male member, and the male member may protrude from the shield. This aspect is advantageous in preventing a drug solution in the flow channel from leaking to the outside in a state in which the male connector is not connected to the female connector.

In an aspect of the connecting device of the present invention above, the male connector may further include, on a side opposite to the male member, a connecting cylinder that is disposed coaxially with the male member and is in communication with the male member. A male thread may be formed on an outer circumferential surface of the connecting cylinder. According to this aspect, the male connector can be connected to, for example, a barrel end of a syringe by means of thread-lock connection. Even if a rotational force is applied to the female connector main body, the rotational force does not loosen the screwing of the thread-lock connection.

Hereinafter, the present invention will be described in detail while showing a preferred embodiment thereof. However, it goes without saying that the present invention is not limited to the embodiment below. In the drawings that will be referred to in the following description, only the main constituent members of the embodiment of the present invention are shown in a simplified manner for the sake of convenience of description. Therefore, the present invention



may include optional members that are not shown in the drawings below. Also, the members shown in the drawings below may be changed or omitted within the scope of the present invention. Identical members are denoted by identical reference numerals in different drawings.

#### 1. Male Connector

A male connector **100** constituting a connecting device according to an embodiment of the present invention will be described. FIG. 1A is a perspective view of the male connector **100**. FIG. 1B is a cross-sectional view of the male connector **100**. FIG. 1C is a cross-sectional perspective view of the male connector **100**. The cross section shown in FIG. 1B and the cross section shown in FIG. 1C are orthogonal to each other, where the central axis (not shown) of the male connector **100** is the line of intersection. FIG. 1D is a plan view of the male connector **100**. The male connector **100** includes a male connector main body **101** and a shield **150**. The male connector main body **101** includes a male member (male luer) **110**, a hood **120**, a lock lever **130**, and a connecting cylinder **140**.

For convenience of the following description, a longitudinal direction of the male member **110** is referred to as “vertical direction”, a direction that is parallel to a plane perpendicular to the longitudinal direction of the male member **110** is referred to as “horizontal direction”. However, the “vertical direction” and the “horizontal direction” do not mean the orientation of the male connector **100** when in actual use. A direction of rotation around the central axis of the male connector **100**, which is also the central axis of the male member **110**, is referred to as “rotating direction” or “circumferential direction”. Furthermore, a direction of a straight line orthogonal to the central axis is referred to as “radial direction”. In the radial direction, the side nearer to the central axis is referred to as “inner side”, and the side farther from the central axis is referred to as “outer side”.

As shown in FIG. 1B, the male member **110** is a straight rod-shaped member. An outer circumferential surface (lateral surface) of the male member **110** is a tapered surface (conical surface) whose outer diameter gradually decreases toward a distal end **111** of the male member **110**. However, the shape of the outer circumferential surface of the male member **110** is not limited to the above-described shape, and any shape can be chosen. For example, the outer circumferential surface of the male member **110** may constitute a cylindrical surface whose outer diameter is constant from a base end to the distal end **111** of the male member **110**. Alternatively, the outer circumferential surface of the male member **110** may be formed of a combination of a tapered surface and a cylindrical surface, or may be formed of any desired curved surface.

A flow channel **112** is formed within the male member **110** and extends in the longitudinal direction of the male member **110**. The flow channel **112** is not open in the distal end **111** of the male member **110**. Two lateral holes **113** that are in communication with the flow channel **112** are formed in the outer circumferential surface of the male member **110** at respective positions near the distal end **111**. Each lateral hole **113** extends in the radial direction of the male member **110** and is open in the outer circumferential surface of the male member **110**. In the present embodiment, the two lateral holes **113** are arranged in a single straight line, but the number of lateral holes **113** is not necessarily required to be two and may also be one or three or more.

A flange **115** extends outward in the radial direction from the base end of the male member **110**. The flange **115** is a substantially circular plate-like member. The hood **120** and

the lock lever **130** extend toward the same side as the male member **110** from an outer circumferential edge of the flange **115**.

The hood **120** has a hollow cylindrical shape. The hood **120** is coaxial with the male member **110**, surrounds the male member **110**, and is spaced apart from the male member **110** in the radial direction. The hood **120** is open upward (toward a side opposite to the flange **115**) at a distal end **121** thereof. The distal end **121** of the hood **120** is located farther from the connecting cylinder **140** than the distal end **111** of the male member **110** in the vertical direction (or in other words, nearer to a distal end of the male connector **100** than the distal end **111**).

An inner circumferential surface (surface that opposes the male member **110**) of the hood **120** is a cylindrical surface whose inner diameter is constant in the longitudinal direction of the male member **110**. The inner diameter of the inner circumferential surface of the hood **120** is slightly larger than the outer diameter of a female connector **200** (in particular, a rotating cylinder **250** thereof, see FIG. 2A, which will be described later) to which the male connector **100** is to be connected. Four ribs (projections) **127** protrude from the inner circumferential surface of the hood **120** toward the male member **110**. The four ribs **127** are spaced apart from each other in the circumferential direction. The ribs **127** extend parallel to the longitudinal direction of the male member **110**, from the distal end **121**, or a position near the distal end **121**, of the hood **120** to the flange **115**. Upper ends of the ribs **127** are located farther from the connecting cylinder **140** than the distal end **111** of the male member **110** in the vertical direction (or in other words, nearer to the distal end of the male connector **100** than the distal end **111**).

The lock lever **130** opposes the male member **110** in the radial direction. The lock lever **130** has an elongated thin plate-like shape (strip-like shape), and its longitudinal direction is substantially parallel to the longitudinal direction of the male member **110**.

The lock lever **130** has a cantilever structure, with its fixed end (base end) being fixed to the flange **115**. A claw **132** protruding toward the male member **110** is provided at or near a free end (distal end) of the lock lever **130**. The claw **132** has an inclined surface **132a** above (on the side opposite to the flange **115**) its protruding end (a portion closest to the male member **110**). The inclined surface **132a** is inclined so as to be closer to the flange **115** toward the protruding end of the claw **132**. The claw **132** protrudes toward the male member **110** than the inner circumferential surface of the hood **120**.

An operating arm **135** protrudes outward (toward a side away from the male member **110**) from a surface of the lock lever **130** that is located on the side opposite to the male member **110**. The operating arm **135** extends beyond the fixed end of the lock lever **130** (or the flange **115**) toward the connecting cylinder **140** from a position between the fixed end and the free end of the lock lever **130** while being spaced apart from the lock lever **130**. An operating portion **136** is provided at a distal end of the operating arm **135**. The operating arm **135** has a mechanical strength that is high enough for the operating arm **135** to be considered as a substantially rigid body.

The lock lever **130** can elastically bend and deform and can thus swing in a plane containing the central axis of the male member **110**. For example, when a force **F** acting inward in the radial direction is applied to the operating portion **136**, the claw **132** moves in the direction of arrow **A** (outward in the radial direction) so as to move away from the male member **110** substantially in the radial direction. A

portion of the lock lever **130** that is located between the fixed end and the position to which the operating arm **135** is connected is an elastic portion **131**. When the force **F** is applied to the operating portion **136**, mainly the elastic portion **131** elastically bends and deforms.

An opening **123** penetrating the hood **120** in the radial direction is formed in the hood **120**. The opening **123** has an inverted substantially “U” shape. The opening **123** does not reach the distal end **121** of the hood **120**. The opening **123** allows the lock lever **130** to be spaced apart from the hood **120** so that the lock lever **130** can swing. The lock lever **130** is disposed within the opening **123** while extending substantially along the cylindrical surface formed by the hood **120**.

The connecting cylinder **140** extends downward (toward a side opposite to the male member **110**) from the flange **115**. The connecting cylinder **140** has a substantially cylindrical shape that is coaxial with the male member **110**, and is in communication with the flow channel **112** of the male member **110**. An inner circumferential surface **141** of the connecting cylinder **140** constitutes a female tapered surface (e.g., a 6% tapered surface) whose inner diameter increases as the distance from the flange **115** increases. A male thread **142** is formed on an outer circumferential surface of the connecting cylinder **140**. The connecting cylinder **140** and the male thread **142** may comply with ISO 594-2, for example. The connecting cylinder **140** including the inner circumferential surface **141** and the male thread **142** constitutes a base end portion of the male connector **100**.

It is preferable that the male connector main body **101** is made of a hard material. Specifically, a resin material such as polyacetal, polycarbonate, polystyrene, polyamide, polypropylene, or rigid polyvinyl chloride can be used. The entire male connector main body **101** can be integrally produced as a single component through injection molding or the like using such a resin material.

The shield **150** is disposed inside the hood **120** so as to cover the male member **110**. The shield **150** includes a head portion **151**, a circumferential wall **155**, and a base portion **158** in this order. The shield **150** is made of an elastic (or flexible) soft material (so-called elastomer) so that it can be relatively easily deformed by an external force and immediately recover to a state before deformation (initial state) when the external force is removed. Examples of soft materials that can be used include, but are not limited to, soft polyvinyl chloride, thermoplastic elastomers (e.g., styrene-based elastomers, olefin-based elastomers, polyurethane-based elastomers, etc.), and rubbers (e.g., isoprene rubber, silicone rubber, butyl rubber, etc.). The entire shield **150** can be integrally produced as a single component using such a material.

The circumferential wall **155** has a hollow cylindrical shape as a whole. When a compressive force acting in the vertical direction is applied to the shield **150**, the circumferential wall **155** is elastically compressively deformed such that its vertical length is reduced (see FIG. 4B, which will be described later), and when released from the compressive force, the circumferential wall **155** immediately returns to the initial state (see Patent Document 1, for example).

The head portion **151** has an inner cavity **152** that is in communication with an inner space of the circumferential wall **155**. It is preferable that an inner circumferential surface of the inner cavity **152** has a shape that conforms to the outer circumferential surface of the male member **110** so as to come into intimate contact with the outer circumferential surface of the male member **110**. In the present

embodiment, the inner circumferential surface of the inner cavity **152** constitutes a cylindrical surface whose inner diameter is constant in the vertical direction. A slit **153** penetrating the head portion **151** in the vertical direction is formed at the innermost portion of the inner cavity **152**. The slit **153** is a straight line-shaped cut portion having a “-” (minus sign) shape in a plan view. A conical surface-shaped (or a mushroom- or umbrella-shaped) convex portion **154** protrudes upward from an upper surface of the head portion **151**. The slit **153** passes through the central apex of the convex portion **154**.

The base portion **158** protrudes outward in the radial direction from a lower end of the circumferential wall **155**. The base portion **158** is placed on the flange **115**, and an annular fixing ring (not shown) is put on the base portion **158**. The fixing ring is locked to the male connector main body **101**. In this manner, the shield **150** is fixed to the male connector main body **101**. However, the method for fixing the shield **150** to the male connector main body **101** is not limited to this method, and any methods can be used. The configuration of the base portion **158** may be changed as appropriate depending on the fixing method.

The distal end **111** and its neighboring portion of the male member **110** are inserted into the inner cavity **152** of the shield **150**. The inner circumferential surface of the inner cavity **152** is expanded in diameter by the male member **110** and is in intimate contact with the male member **110**. The distal end **111** of the male member **110** is spaced apart from the slit **153** of the shield **150** in the vertical direction. A liquid-tight space defined by the distal end **111** and the inner circumferential surface of the inner cavity **152** is formed. As shown in FIG. 1C, the lateral holes **113** of the male member **110** are closed by the inner circumferential surface of the inner cavity **152** in a liquid-tight manner.

FIGS. 1A to 1D show the male connector **100** in the initial state. In the initial state, the shield **150** is not substantially compressively deformed in the vertical direction, the slit **153** is closed in a liquid-tight manner, and the lock lever **130** is not bent and deformed.

## 2. Female Connector

The female connector **200** constituting the connecting device according to the embodiment of the present invention will be described. FIG. 2A is a perspective view of the female connector **200**. FIG. 2B is a plan view of the female connector **200**. FIG. 2C is a cross-sectional view of the female connector **200**. The cross section shown in FIG. 2C is taken along line 2C-2C in FIG. 2B, and this cross section contains the central axis (not shown) of the female connector **200**. FIG. 2D is an exploded perspective view of the female connector **200**. For convenience of the following description, a direction that is parallel to the central axis of the female connector **200** is referred to as “vertical direction”, and a direction that is parallel to a plane perpendicular to the central axis of the female member **200** is referred to as “horizontal direction”. However, the “vertical direction” and the “horizontal direction” do not mean the orientation of the female connector **200** when in actual use. A direction of rotation around the central axis is referred to as “rotating direction” or “circumferential direction”. Furthermore, a direction of a straight line orthogonal to the central axis of the female connector **200** is referred to as “radial direction”. In the radial direction, the side nearer to the central axis is referred to as “inner side”, and the side farther from the central axis is referred to as “outer side”.

As shown in FIG. 2D, the female connector **200** includes a female connector main body **210** and a rotating cylinder **250**. As shown in FIG. 2C, the female connector main body

**210** includes a partition member (hereinafter referred to as “septum”) **220** and a housing **230**. The housing **230** includes a housing base **231** and a cap **240**.

The septum **220** has a circular thin plate-like shape. A straight-line shaped slit (cut) **223** penetrating the septum **220** in its thickness direction is formed at the center of the septum **220**. The septum **220** is made of an elastic (or flexible) soft material (so-called elastomer) so that it can be relatively easily deformed by an external force and immediately return to a state before deformation (initial state) when the external force is removed. Examples of soft materials that can be used include, but are not limited to, soft polyvinyl chloride, thermoplastic elastomers (e.g., styrene-based elastomers, olefin-based elastomers, polyurethane-based elastomers, etc.), and rubbers (e.g., isoprene rubber, silicone rubber, butyl rubber, etc.). The entire septum **220** can be integrally produced as a single component using such a material. FIGS. 2A to 2D show the female connector **200** in the initial state. In the initial state, the slit **223** is closed in a liquid-tight manner. The female connector **200** including the septum **220** in which the slit **223** is formed is generally called a needleless port.

The housing base **231** has a substantially hollow cylindrical shape as a whole, and a flow channel **232** penetrates the housing base **231** along the central axis. An upper section of the housing base **231** has a double cylinder structure consisting of an inner cylinder **233** that defines the flow channel **232** and an outer cylinder (first outer cylinder) **234** that is spaced apart from the inner cylinder **233** in a radially outward direction. The inner cylinder **233** and the outer cylinder **234** both have a substantially cylindrical shape and are arranged coaxially. The septum **220** is placed on a distal end of the inner cylinder **233** so as to close the flow channel **232**. The cap **240** is put on the septum **220**. A locking cylinder **244** extending from a top plate **241** of the cap **240** is inserted between the inner cylinder **233** and the outer cylinder **234** and fixed to the outer cylinder **234**. Thus, the septum **220** is sandwiched between the housing base **231** (in particular, the inner cylinder **233** thereof) and the cap **240** (in particular, the top plate **241** thereof) in the vertical direction. A circular opening **242** is formed at the center of the circular top plate **241** of the cap **240**. A region of the septum **220** where the slit **223** is formed is exposed through the opening **242** (see FIG. 2A).

A lower section of the housing base **231** includes a hollow male luer **237** that defines the flow channel **232** and an outer cylinder (second outer cylinder) **238** that surrounds the male luer **237**. An outer circumferential surface of the male luer **237** constitutes a male tapered surface (e.g., a 6% tapered surface) whose outer diameter decreases as the distance to a distal end of the male luer **237** decreases (that is, as the distance from the septum **220** increases). The outer cylinder **238** has a substantially cylindrical shape and is disposed coaxially with the male luer **237** while being spaced apart from the male luer **237** in the radial direction. A female thread **239** is formed on an inner circumferential surface of the outer cylinder **238**. The male luer **237**, the outer cylinder **238**, and the female thread **239** may comply with ISO 594-2, for example. Note that the configuration of the lower section of the housing base **231** is not limited to this configuration, and any configuration may be adopted. The lower section of the housing base **231** constitutes a base end portion of the female connector **200** (female connector main body **210**).

As shown in FIG. 2D, an outer circumferential surface of the outer cylinder **234** includes a cylindrical surface **235**. An annular groove **236** is formed below and adjacent to the cylindrical surface **235**. The annular groove **236** is continu-

ous in the circumferential direction over the entire circumference of the housing base **231**.

It is preferable that the housing base **231** and the cap **240** are made of a hard material. Specifically, a resin material such as polyacetal, polycarbonate, polystyrene, polyamide, polypropylene, or rigid polyvinyl chloride can be used. The entire housing base **231** and the entire cap **240** can each be integrally produced as a single component through injection molding or the like using such a resin material.

The rotating cylinder **250** has a hollow substantially cylindrical shape. An annular projection **256** protrudes inward from an inner circumferential surface of the rotating cylinder **250**. The annular projection **256** extends in the circumferential direction. The inner circumferential surface of the rotating cylinder **250** includes a cylindrical surface **255** above and adjacent to the annular projection **256**. The inner diameter of the rotating cylinder **250** at the position of the annular projection **256** is smaller than the outer diameter of the cylindrical surface **235** of the female connector main body **210**. The inner diameter of the cylindrical surface **255** is slightly larger than the outer diameter of the cylindrical surface **235**.

Four grooves (or slits) **257** are formed in an outer circumferential surface of the rotating cylinder **250**. The four grooves **257** are spaced apart from each other in the circumferential direction. The grooves **257** extend downward from a distal end of the rotating cylinder **250** (upper end of the rotating cylinder **250** in FIG. 2D), are parallel to the central axis of the rotating cylinder **250**, and have a predetermined length.

An opening **252** penetrating the rotating cylinder **250** in the radial direction is formed in the rotating cylinder **250**. The opening **252** is disposed in a region where no grooves **257** are formed. The opening **252** has a substantially rectangular opening shape. A guide groove **253** is formed in the outer circumferential surface of the rotating cylinder **250**. The guide groove **253** extends from the upper end of the rotating cylinder **250** to the opening **252**. The width (circumferential length) of the guide groove **253** is constant in the vertical direction and is equal to the circumferential length of the opening **252**.

The female connector main body **210** is inserted into the rotating cylinder **250** from below the rotating cylinder **250** (see FIG. 2D). A slit **259** (see FIGS. 2D and 3B) penetrating the rotating cylinder **250** in the radial direction extends from a lower end of the rotating cylinder **250** to a position higher than the annular projection **256**. The slit **259** makes it easy for the rotating cylinder **250** to expand in diameter when the annular projection **256** passes over the cylindrical surface **235**.

As shown in FIG. 2C, when the rotating cylinder **250** is attached to the female connector main body **210**, the annular projection **256** of the rotating cylinder **250** fits into the annular groove **236** of the female connector main body **210**. The rotating cylinder **250** is positioned in the vertical direction relative to the female connector main body **210**. The distal end of the rotating cylinder **250** and the top plate **241** of the cap **240** are substantially coplanar with each other.

Once the annular projection **256** fits into the annular groove **236**, it is difficult to separate the rotating cylinder **250** from the female connector main body **210**. In a state in which the annular projection **256** is fitted in the annular groove **236**, the rotating cylinder **250** can freely rotate around the female connector main body **210**. The cylindrical surface **255** of the rotating cylinder **250** faces the cylindrical

surface **235** of the female connector main body **210** in the radial direction, and slides on the cylindrical surface **235**.

It is preferable that the rotating cylinder **250** is made of a hard material. Specifically, resin materials such as polyethylene, polypropylene, polycarbonate, styrene-ethylene, polyethylene terephthalate, polybutylene terephthalate, and a butylene-styrene block copolymer can be used. When consideration is given to the ease of deformation of the rotating cylinder **250** during attachment of the rotating cylinder **250** to the female connector main body **210** and the durability thereof against engagement of the claw **132** of the lock lever **130** with the rotating cylinder **250** (see FIG. 4B), it is preferable that the material of the rotating cylinder **250** has toughness, and in view of this, a polyolefin-based resin such as polyethylene or polypropylene is preferable. The entire rotating cylinder **250** can be integrally produced as a single component through injection molding or the like using such a resin material.

### 3. Connecting Device

A connecting device **1** according to the embodiment of the present invention is constituted by the male connector **100** (see FIGS. 1A to 1D) and the female connector **200** (see FIGS. 2A to 2D). The connecting device **1** can be used to administer a drug solution to a patient. A method of using the connecting device **1** will be described below.

FIG. 3A is a perspective view illustrating a method of using the connecting device **1**. FIG. 3B is a cross-sectional view of main portions in FIG. 3A. The cross section shown in FIG. 3B is taken along the central axis (not shown) shared by the male connector **100** and the female connector **200** and the lock lever **130**.

The male connector **100** is connected to a barrel end **182** of an outer barrel **181** of a syringe **180**. As shown in FIG. 3B, the barrel end **182** includes a male luer **183** and a lock cylinder **184** surrounding the male luer **183**. The male luer **183** is in communication with an inner cavity of the syringe **180** (or the outer barrel **181**). An outer circumferential surface of the male luer **183** constitutes a male tapered surface (e.g., a 6% tapered surface) whose outer diameter gradually decreases toward its distal end. A female thread **185** is formed on an inner circumferential surface of the lock cylinder **184**. The connecting cylinder **140** of the male connector **100** is inserted into a gap between the male luer **183** and the lock cylinder **184**. The inner circumferential surface (female tapered surface) **141** of the connecting cylinder **140** and the male tapered surface of the male luer **183** are taper-fitted to each other, and the male thread **142** of the connecting cylinder **140** is screwed into the female thread **185** of the lock cylinder **184**. The connecting cylinder **140** and the barrel end **182** are securely fitted to each other by means of so-called thread-lock connection. The connection between the connecting cylinder **140** and the barrel end **182** may comply with ISO 594-2. A drug solution is stored in the syringe **180**.

The female connector **200** is provided at an upstream end of a catheter (not shown). The catheter is inserted into a patient, and the distal end of the catheter reaches a desired site (e.g., a vein, bladder, medullary cavity, or gastrointestinal tract). In the present embodiment, the base end portion (the male luer **237** and the outer cylinder **238**) of the female connector **200** (the female connector main body **210**) is connected to a connector (not shown) provided at the upstream end of the catheter. The connector of the catheter has substantially the same configuration as the connecting cylinder **140**, and this connector and the female connector **200** are connected to each other in compliance with ISO 594-2. However, the configuration of the connection portion

between the catheter and the female connector **200** is not limited to that of the present embodiment, and may be changed as desired. The base end portion of the female connector **200** may be changed as appropriate depending on the configuration of the connector provided at the upstream end of the catheter.

As shown in FIG. 3A, the distal end of the male connector **100** is placed coaxially with and opposite to the distal end of the female connector **200**. A worker can hold the outer barrel **181** or the hood **120** with one hand (e.g., right hand) and the rotating cylinder **250** with the other hand (e.g., left hand). The male connector **100** and the rotating cylinder **250** are aligned with each other in the circumferential direction such that the circumferential position of the guide groove **253** and the opening **252** of the rotating cylinder **250** coincides with that of the lock lever **130**. Since the rotating cylinder **250** can freely rotate relative to the female connector main body **210**, the catheter connected to the female connector main body **200** does not twist even when the rotating cylinder **250** is rotated. Therefore, it is easy to align the male connector **100** and the rotating cylinder **250** with each other in the circumferential direction.

Then, the female connector **200** is inserted and pushed into the hood **120** of the male connector **100**.

As can be seen from FIG. 3B, the female connector **200** (in particular, the top plate **241** and the septum **220**) collides with the convex portion **154** of the shield **150** and moves the head portion **151** toward the syringe **180**. The male member **110** passes through the slit **153** of the shield **150** and further enters the slit **223** of the septum **220**. In parallel with this, a distal end **253a** (see FIG. 2A) of the guide groove **253** of the rotating cylinder **250** comes into contact with the inclined surface **132a** (see FIGS. 1B and 1C) of the claw **132** of the lock lever **130**. While sliding on the inclined surface **132a**, the distal end **253a** causes the lock lever **130** to elastically bend and deform in a direction (direction indicated by arrow A in FIG. 1B) in which the claw **132** moves away from the male member **110**. Then, the claw **132** slides on the guide groove **253**. When the claw **132** finishes passing through the guide groove **253**, the lock lever **130** elastically recovers, and the claw **132** fits into the opening **252**.

FIG. 4A is a perspective view of the connecting device **1** in which the male connector **100** is connected to the female connector **200**. FIG. 4B is a cross-sectional view of main portions in FIG. 4A. The male member **110** passes through the slit **153** (see FIG. 1B) of the shield **150** and the slit **223** (see FIG. 2C) of the septum **220** in this order. The convex portion **154** (see FIG. 1B) of the shield **150** enters the inside of the opening **242** (see FIG. 2C) of the cap **240**. The septum **220** is significantly deformed into the flow channel **232** by the male member **110** and the shield **150**. The lateral holes **113** of the male member **110** are exposed in the flow channel **232**. The flow channel **112** of the male member **110** and the flow channel **232** of the female connector **200** are in communication with each other. The shield **150** receives a compressive force acting in the vertical direction, and the circumferential wall **155**, in particular, is compressively deformed such that the vertical length of the shield **150** is reduced. The lock lever **130** has recovered to substantially the same state as the initial state. The claw **132** of the lock lever **130** is engaged with the rotating cylinder **250** (locked state). Even if a pulling force is applied between the male connector **100** and the female connector **200**, the male connector **100** cannot be disconnected from the female connector **200**.

The four ribs **127** (see FIG. 1D) of the male connector **100** and the four grooves **257** (see FIG. 2B) of the rotating

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cylinder **250** are arranged such that, when the lock lever **130** (in particular, the claw **132** thereof) is aligned with the guide groove **253** and the opening **252** of the rotating cylinder **250** in the circumferential direction as shown in FIG. 3A, the circumferential positions of the four ribs **127** coincide with the circumferential positions of the four grooves **257**. Therefore, although not shown, the ribs **127** fit into the grooves **257** when the male connector **100** is connected to the female connector **200** (see FIGS. 4A and 4B).

In the state in which the male connector **100** is connected to the female connector **200** (see FIGS. 4A and 4B), a plunger **188** of the syringe **180** is pushed in to administer the drug solution in the syringe **180** to the patient through the male connector **100** and the female connector **200**.

To disconnect the male connector **100** from the female connector **200**, the operating portion **136** of the lock lever **130** is pushed inward in the radial direction (direction indicated by arrow F in FIG. 1B). The lock lever **130** elastically bends and deforms, and the claw **132** moves outward in the radial direction. The claw **132** comes out of the opening **252**, and the engagement of the claw **132** with the rotating cylinder **250** is cancelled (unlocked state). In this state, if the male connector **100** and the female connector **200** are pulled away from each other, the male connector **100** can be disconnected from the female connector **200**. The septum **220** elastically recovers as soon as the male member **110** is removed therefrom, the slit **223** is closed in a liquid-tight manner. The shield **150** expands due to the elastic recovery force it has. The male member **110** is accommodated in the shield **150**, and the lateral holes **113** are closed by the shield **150**. The slit **153** is closed in a liquid-tight manner as soon as the male member **110** is removed therefrom. If the force applied to the operating portion **136** is released, the lock lever **130** elastically recovers. The male connector **100** and the female connector **200** each return to the initial state (see FIGS. 1A to 1D, 2A to 2D, 3A, and 3B).

In general, in a hospital where a plurality of patients are hospitalized, a pharmacist puts a drug solution prepared for each patient's site of administration into a syringe by suction. A nurse then administers the drug solution in the syringe to the patient through a catheter placed in the patient. In this case, if the syringe can be connected to any catheter, a misconnection accident may occur in which, for example, the nurse mistakenly connects the syringe storing a drug solution prepared for intravenous administration to a catheter placed in the bladder.

The connecting device **1** of the present embodiment can prevent such accidents. This will be described below.

As described above, the four ribs **127** are formed on the inner circumferential surface of the hood **120** of the male connector **100**. The four grooves **257** are formed in the outer circumferential surface of the rotating cylinder **250** of the female connector **200**. When the female connector **200** is inserted into the hood **120**, the four ribs **127** fit into the four grooves **257** respectively. Thus, the male connector **100** can be connected to the female connector **200**. If the rotating cylinder **250** does not have the grooves **257** corresponding to the ribs **127** of the hood **120**, the female connector **200** cannot be inserted into the hood **120**, and therefore, the male connector **100** cannot be connected to the female connector **200**.

In the present embodiment, the four grooves **257** are formed in the rotating cylinder **250**. In the present embodiment, for example, if at least one groove **257** of the four grooves **257** is omitted, the circumferential position of at least one groove **257** of the four grooves **257** is changed, the

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width of at least one groove **257** of the four grooves **257** is reduced, or the depth of at least one groove **257** of the four grooves **257** is reduced, then, the male connector **100** cannot be connected to the female connector **200**.

Also, in the present embodiment, the four ribs **127** are formed in the hood **120**. In the present embodiment, for example, if at least one rib **127** is added, the circumferential position of at least one rib **127** of the four ribs **127** is changed, the width of at least one rib **127** of the four ribs **127** is increased, or the protruding height of at least one rib **127** of the four ribs **127** is increased, then, the male connector **100** cannot be connected to the female connector **200**.

Therefore, the number, circumferential position, dimensions (width and protruding height), and the like of ribs **127** formed in the hood **120** as well as the number, circumferential position, dimensions (width and depth), and the like of grooves **257** formed in the rotating cylinder **250** are standardized for each site (e.g., a vein, bladder, medullary cavity, gastrointestinal tract, or the like) to which the drug solution is to be administered. That is to say, connecting devices **1** that comply with a plurality of standards with different numbers, circumferential positions, dimensions, and the like of ribs **127** and grooves **257** are prepared for different sites to which the drug solution is to be administered. This allows a male connector to be selectively connected only to a female connector that complies with the same standard as the male connector. It is not possible to connect a male connector **100** that complies with a certain standard (e.g., a standard for intravenous administration) to a female connector **200** that complies with a different standard (e.g., a standard for bladder). As a result, the above-described misconnection accidents can be prevented.

It is relatively easy to produce a plurality of types of male connectors **100** that differ only in the number, circumferential position, dimensions, and the like of ribs **127** and to produce a plurality of types of female connectors **200** that differ only in the number, circumferential position, dimensions, and the like of grooves **257**. Therefore, according to the present embodiment, misconnection accidents can be prevented with ease and at low cost.

It is sufficient that the rib(s) **127** and the groove(s) **257** are configured so as to allow a male connector and a female connector that comply with the same standard to be connected to each other but not allow a male connector and a female connector that comply with different standards to be connected to each other. The features of the rib(s) **127** and the groove(s) **257** that enable such selective connection are not limited to the number, circumferential position, and dimensions thereof. For example, the rib(s) **127** and the groove(s) **257** may have different cross-sectional shapes (shapes in cross sections along a horizontal plane) for different standards. Therefore, the features of the rib(s) **127** and the groove(s) **257** specified in each standard are not limited to the number, circumferential position, and dimensions, and any features can be specified. The number, circumferential positions, and dimensions of the ribs **127** and the grooves **257** shown in the present embodiment are merely examples, and any changes can be made thereto within the scope of the present invention. In general, it is preferable to realize selective connection between male connectors and female connectors by varying the number of ribs **127** and grooves **257**. The reasons for this are that the worker can easily notice the difference in the number of ribs **127** and grooves **257**, and it is relatively easy to produce male connectors **100** and female connectors **200** with different numbers of ribs **127** and grooves **257**.

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In the present embodiment, the ribs **127** extend from the distal end **121**, or a position near the distal end **121**, of the hood **120**, and upper ends (starting ends) of the ribs **127** are located farther from the connecting cylinder **140** than the distal end **111** of the male member **110** in the vertical direction (or in other words, nearer to the distal end of the male connector **100** than the distal end **111**) (see FIGS. **1A** **1C**). Moreover, the grooves **257** extend from the distal end (the position of the top plate **241**), or a position near the distal end, of the female connector **200** (see FIG. **2A**). Therefore, if an attempt is made to connect a male connector **100** to a female connector **200** that complies with a different standard than the standard with which this male connector **100** complies (that is, if a misconnection is attempted), the ribs **127** cannot be fitted into the grooves **257**, and thus the female connector **200** cannot be substantially inserted into the hood **120**. Unlike the present embodiment, if the ribs **127** extend from positions retracted from the distal end **121** of the hood **120** and/or the grooves **257** extend from positions away from the distal end of the female connector **200**, it is possible to insert a portion of the female connector **200** into the hood **120** even when the ribs **127** cannot be fitted into the grooves **257**. In this case, the convex portion **154** of the shield **150** may come into contact with the septum **220**, causing the slit **153** of the shield **150** and/or the slit **223** of the septum **220** to open, and the drug solution may thus leak to the outside. In the case where the drug solution contains a hazardous drug, the worker will be exposed to the drug. The present embodiment can prevent these situations.

The ribs **127** extend in the direction of the central axis of the male connector **100**, and the grooves **257** extend in the direction of the central axis of the female connector **200**. Therefore, in the process of connecting the male connector **100** to the female connector **200** and in the process of disconnecting the male connector **100** from the female connector **200**, while the ribs **127** are fitted in the grooves **257**, the male connector **100** can move in the central axis direction (i.e., the longitudinal direction of the male member **110**) relative to the female connector **200** (the female connector main body **210** and the rotating cylinder **250**). Thus, the male connector **100** can be connected to and disconnected from the female connector **200** by simply moving (translating) the male connector **100** in the central axis direction relative to the female connector **200** (the female connector main body **210** and the rotating cylinder **250**). This is advantageous in improving the ease of connecting and disconnecting operations.

The female connector **200** includes the rotating cylinder **250** that is rotatable around the female connector main body **210**, and the grooves **257** and the opening **252** are formed in the rotating cylinder **250**. Therefore, in a state in which the male connector **100** is connected to the female connector **200** (see FIGS. **4A** and **4B**), or in other words, in a state in which the ribs **127** are fitted in the grooves **257** and the claw **132** of the lock lever **130** is fitted in the opening **252**, transmission of a rotational force between the male connector **100** and the female connector main body **210** is blocked. For example, even if the catheter connected to the female connector main body **210** is twisted, the twisting does not loosen the screwing (see FIG. **4B**) of the thread-lock connection between the connecting cylinder **140** and the barrel end **182**. Moreover, even if the male connector **100** and the syringe **180** are rotated, the catheter connected to the female connector main body **210** does not twist, and loosening of the screwing of the thread-lock connection between the connecting cylinder **140** and the barrel end **182** caused by a torsional reaction force of the catheter does not occur.

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The drug solution may possibly leak to the outside if the screwing of the thread-lock connection loosens. In the case where the drug solution contains a hazardous drug, the worker will be exposed to the drug. The present embodiment can prevent these situations.

The male connector **100** includes the lock lever **130** provided with the claw **132**. The state in which the male connector **100** is connected to the female connector **200** is stably maintained by the engagement of the claw **132** with the female connector **200**. The lock lever **130** with the claw **132** is advantageous in preventing the occurrence of unintentional disconnection of the male connector **100** from the female connector **200** and a resulting leakage of the drug solution from the male connector **100** and/or the female connector **200**. In the case where the drug solution contains a hazardous drug, exposure of the worker to the drug can be prevented.

The claw **132** of the lock lever **130** engages with the rotating cylinder **250** rather than the female connector main body **210**. Thus, as described above, in a state in which the claw **132** is engaged with the rotating cylinder **250**, transmission of the rotational force between the male connector **100** and the female connector main body **210** is blocked.

The male connector **100** includes the shield **150** that closes the lateral holes **113** of the male member **110**. Therefore, like a conventional male connector (see Patent Document 1), the male connector **100** can prevent leakage of the drug solution in the flow channel **112** to the outside when the female connector **200** is not connected thereto. The female connector **200** includes the septum **220** in which the slit **223** is formed. Therefore, like a conventional female connector (see Patent Document 1), the female connector **200** can prevent leakage of the drug solution in the flow channel **232** to the outside when the male connector **100** is not connected thereto. Accordingly, the connecting device **1** including the male connector **100** and the female connector **200** can be used as a closed-system connecting device. In a series of operations for transferring the drug solution in the syringe **180** to the catheter connected to the female connector **200**, leakage of the drug solution to the outside is prevented. Even in the case where the drug solution contains a hazardous drug, the worker will not be exposed to the drug.

#### 4. Various Modifications

The present invention is not limited to the above-described embodiment, and changes can be made thereto as appropriate.

In the above-described embodiment, in order to realize selective connection between male connectors and female connectors, the male connector **100** is provided with the ribs **127**, and the female connector **200** is provided with the grooves **257**. However, the present invention is not limited to this configuration. For example, a configuration may be adopted in which grooves are formed in the inner circumferential surface of the hood **120** of the male connector **100**, and ribs that can fit into the grooves are formed on the outer circumferential surface of the rotating cylinder **250** of the female connector **200**. In general, selective connection can be realized by providing the male connector and the female connector with respective fitting structures that can fit to each other when the male connector is connected to the female connector. It is preferable that, in a state in which a first fitting structure provided on the male connector and a second fitting structure provided on the female connector are fitted to each other, the first fitting structure can move in the central axis direction relative to the second fitting structure. It is preferable that the fitting of the first fitting structure to the second fitting structure is maintained as long as the

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female connector is inserted in the hood of the male connector. One of the first fitting structure and the second fitting structure may be a protrusion (e.g., a rib), and the other of the first fitting structure and the second fitting structure may be a depression (e.g., a groove) into which the protrusion can fit. Such protrusion and depression may have any length in the central axis direction, but it is preferable that at least one of the protrusion and the depression extends in the central axis direction.

In the above-described embodiment, the ribs **127** and the grooves **257** are varied according to the site to which the drug solution is to be administered; however, the present invention is not limited to this configuration. For example, the ribs **127** and the grooves **257** may be varied according to the type of the liquid flowing through the male connector and the female connector.

In the above-described embodiment, the claw **132** of the lock lever **130** engages with the rotating cylinder **250**; however, the present invention is not limited to this configuration. The claw **132** may engage with the female connector main body **210** (the housing base **231**). For example, it is possible to form a step (or a recess) on an exposed portion of the outer circumferential surface of the female connector main body **210** that is located nearer to the base end portion (nearer to the male luer **237**) than the rotating cylinder **250** and cause the claw **132** to engage with this step (or recess). The claw **132** engaging with the female connector main body **210** is advantageous in counteracting a greater pulling force that is applied between the male connector **100** and the female connector **200** in a state in which the male connector **100** is connected to the female connector **200**. The step (or recess) with which the claw **132** engages may be continuous over the entire circumference of the female connector main body **210** in the circumferential direction. In this case, while the claw **132** engages with the female connector main body **210**, the male connector **100** can rotate relative to the female connector main body **210**, and therefore, as in the case of the above-described embodiment, transmission of the rotational force between the male connector **100** and the female connector main body **210** is blocked.

Also, the configuration of the lock lever **130** provided in the male connector **100** is not limited to that shown in the above-described embodiment. Instead of the above-described lock lever **130**, a known lock lever (see Patent Document 1, for example) to be provided in a male connector can be applied to the present invention. The number of lock levers is not limited to one, and may be two or more. When the claw **132** moves in the radial direction, the flange **115** that supports the lock lever **130** may elastically bend and deform, instead of the lock lever **130**.

The shape of the opening at the distal end of the hood **120** is set to such a shape that enables the female connector **200** to be inserted into and removed from the hood **120** and enables the female connector **200** inserted in the hood **120** to be positioned in the horizontal direction. The shape of the opening at the distal end of the hood **120** is not limited to a circle such as that shown in the above-described embodiment. For example, the cross-sectional shape of the outer circumferential surface of the rotating cylinder **250** taken along a horizontal plane may be an ellipse, and, in this case, it is preferable that the shape of the opening at the distal end of the hood **120** is substantially the same ellipse as that of the rotating cylinder **250**.

The cross-sectional shape of the inner circumferential surface of the hood **120** does not need to be constant in the

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central axis direction, and, for example, the inner diameter of the hood **120** may gradually increase toward the flange **115**.

Also, the configuration of the shield **150** provided in the male connector **100** is not limited to that shown in the above-described embodiment. Instead of the self-closing slit **153**, a hole (see Patent Document 1) through which the distal end **111** of the male member **110** is exposed may be formed at the distal end of the shield **150**. In the present invention, the male connector **100** need not include the shield **150**.

The male connector of the present invention may be connected to a member (e.g., a flexible tube and various containers) other than the syringe **180**. Depending on the member to which the male connector is connected, the configuration of the base end portion (connecting cylinder **140**) of the male connector can be changed as appropriate.

Also, the configuration of the female connector is not limited to that shown in the above-described embodiment. For example, the structure (the inner cylinder **233**, the outer cylinder **234**, and the locking cylinder **244**) of a fixed portion of the cap **240** fixed to the housing base **231** can be changed as desired. The cylindrical surface **235** (see FIG. 2D) may be formed by the cap **240**.

The configuration of the rotating cylinder **250** of the female connector can also be changed as desired. For example, instead of the opening **252**, a recess that does not penetrate the rotating cylinder **250** in the radial direction may be formed in the outer circumferential surface of the rotating cylinder **250**. The guide groove **253** may be omitted.

The female connector of the present invention may be connected to a member (e.g., a flexible tube) other than a catheter inserted in a patient. Depending on the member to which the female connector is connected, the configuration of the base end portion (the male luer **237**, the outer cylinder **238**, and the female thread **239**) of the female connector can be changed as appropriate. In addition, the female connector of the present invention does not need to be connected to an end of a tube, and may be provided as a coinjection port in the middle of a tube, for example. Alternatively, the female connector of the present invention may be provided on an adapter (see Patent Document 2, for example) to be attached to a vial.

There is no limitation on the liquid flowing through the connecting device of the present invention. This liquid may be a drug solution containing no hazardous drug, blood, a nutrient, water (including physiological saline solution), various solutions, or the like.

#### INDUSTRIAL APPLICABILITY

The present invention can be widely used to form a flow channel through which a liquid flows. In particular, the present invention can be favorably used when it is necessary to prevent a misconnection between a male connector and a female connector. The present invention can be favorably used in the field of medicine. However, in addition to the field of medicine, the present invention can also be used in other fields in which liquids are handled, such as the fields of food, chemistry, machinery, and the like.

#### LIST OF REFERENCE NUMERALS

- 1 Connecting device
- 100 Male connector
- 110 Male member
- 111 Distal end of male member
- 113 Lateral hole (opening of flow channel)

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- 112 Flow channel of male member
- 120 Hood
- 127 Rib (Projection, First fitting structure)
- 130 Lock lever
- 132 Claw
- 140 Connecting cylinder
- 142 Male thread
- 150 Shield
- 200 Female connector
- 210 Female connector main body
- 220 Septum (partition member)
- 223 Slit
- 250 Rotating cylinder
- 257 Groove (Second fitting structure)

The invention claimed is:

1. A connecting device comprising a male connector and a female connector that can be connected to and disconnected from each other,
  - wherein the male connector includes:
    - a rod-shaped male member;
    - a hood surrounding the male member; and
    - a first fitting structure provided on an inner circumferential surface of the hood,
  - the female connector includes:
    - a female connector main body including a partition member in which a slit is formed, the partition member being made of an elastic material;
    - a rotating cylinder provided on the female connector main body so as to be rotatable around the female connector main body; and
    - a second fitting structure provided on an outer circumferential surface of the rotating cylinder, and
  - when the male connector is connected to the female connector, the male member passes through the slit of the partition member, the female connector is inserted into the hood, and the first fitting structure fits to the second fitting structure.
2. The connecting device according to claim 1, wherein the female connector cannot be inserted into the hood when the first fitting structure does not fit to the second fitting structure.
3. The connecting device according to claim 1, wherein the first fitting structure is provided nearer to a distal end of the male connector than a distal end of the male member.
4. The connecting device according to claim 1, wherein the second fitting structure is provided at or near a distal end of the female connector.

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5. The connecting device according to claim 1, wherein, in a process of connecting the male connector to the female connector and in a process of disconnecting the male connector from the female connector, the male connector is movable in a longitudinal direction of the male member relative to the female connector, with the first fitting structure being fitted to the second fitting structure.
6. The connecting device according to claim 1, wherein one of the first fitting structure and the second fitting structure is a groove extending in a direction in which the male connector is connected to and disconnected from the female connector, and another one of the first fitting structure and the second fitting structure is a projection that can fit into the groove.
7. The connecting device according to claim 1, wherein, in a state in which the male connector is connected to the female connector, the male connector is rotatable relative to the female connector main body.
8. The connecting device according to claim 1, wherein the male connector further includes a lock lever and a claw provided on the lock lever, the claw is elastically displaceable in a direction away from the male member, and when the male connector is connected to the female connector, the claw engages with the female connector.
9. The connecting device according to claim 8, wherein, when the male connector is connected to the female connector, the claw engages with the rotating cylinder.
10. The connecting device according to claim 1, wherein the male connector further includes a shield that is compressively deformable in a longitudinal direction of the male member, in a state in which the male connector is not connected to the female connector, the shield closes an opening of a flow channel provided in the male member, and in a state in which the male connector is connected to the female connector, the shield is compressively deformed in the longitudinal direction of the male member, and the male member protrudes from the shield.
11. The connecting device according to claim 1, wherein the male connector further includes, on a side opposite to the male member, a connecting cylinder that is disposed coaxially with the male member and is in communication with the male member, and a male thread is formed on an outer circumferential surface of the connecting cylinder.

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